WATTWISE AI

Smart Energy Consumption Pattern & Anomaly Discovery

Concept Note | AICTE–IBM SkillsBuild Machine Learning & Applied AI Internship

1. Executive Summary

WattWise AI is a student-built prototype that explores how machine learning can help people review energy-consumption data. It presents hourly readings in a dashboard and applies the unsupervised Isolation Forest algorithm to flag readings that differ from the patterns in the dataset. The intention is to support investigation and energy awareness, not to automatically diagnose faults or claim energy savings.

2. Problem Statement

Energy-consumption data can be difficult to interpret when it is viewed as a long list of readings. Unusual spikes may go unnoticed or require manual spreadsheet review. A lightweight tool that visualizes usage and highlights records for human review could make basic pattern discovery more accessible to households, campus facilities, and student sustainability teams.

3. Objectives

• Build a responsive dashboard for timestamped energy readings.

• Apply an unsupervised ML method to flag potentially unusual records.

• Allow CSV upload and provide a downloadable file containing model flags and anomaly scores.

• Explain model output and limitations clearly so that users do not treat flags as confirmed faults.

• Create a low-cost prototype that can be evaluated later with consented, real-world data.

4. Proposed Solution and Workflow

The user loads a CSV containing timestamp, hour, day_of_week, temperature_c, and consumption_kwh. The application validates the columns, converts numeric fields, and removes rows with invalid feature values. An Isolation Forest model is fit to the dataset being analyzed, using time and temperature context alongside consumption. The dashboard displays total readings, flagged readings, average and maximum consumption, a recent timeline, and a table of recent model outputs. Users can download scored sample records.

5. Technology Stack

Layer	Technology	Purpose
Language	Python	Application logic and data processing
Web framework	Flask	Routes, file upload and dashboard rendering
Data	Pandas, NumPy	CSV handling and numeric transformations
Machine learning	scikit-learn IsolationForest	Unsupervised outlier detection
Frontend	HTML, CSS, Chart.js	Responsive interface and consumption chart

6. SDG Alignment and Expected Impact

SDG 7 — Affordable and Clean Energy: supports energy awareness and the exploration of consumption patterns. **SDG 12 — Responsible Consumption and Production:** encourages review of usage patterns that may

warrant investigation. The prototype's impact is currently prospective; no reduction in consumption, emissions, or cost has been measured.

7. Scope, Dataset, and Evaluation Plan

The included dataset is synthetic and spans 30 days of hourly readings, with several deliberately elevated readings to make a demonstration possible. The model is fit to the dataset being analyzed and uses a configured contamination fraction of 2.5% as an initial demo setting. This parameter is not calibrated to a utility or building. A future pilot should use authorized meter data, compare flagged readings with verified events and simple baselines, and measure precision, false-positive rate, and investigation time.

8. Limitations, Ethics, and Risk Controls

A flagged reading is an outlier relative to the supplied data; it is not proof of waste, equipment failure, billing error, or a safety concern. Results can be affected by seasonality, sensor quality, changes in occupancy, and dataset composition. The prototype does not connect to meters or utility APIs. Users should avoid uploading personal or sensitive data without authorization. Public deployment would require authentication, access control, secure handling, and privacy review.

9. Future Enhancements

- Validate on consented, real energy data and document data quality.
- Add baseline comparisons and allow experts to review or label alerts.
- Explore seasonal profiles and explainable anomaly summaries.
- Add optional facility-level reports and privacy-conscious deployment after evaluation.

10. Conclusion

WattWise AI demonstrates a practical application of data processing, unsupervised machine learning, and web development for energy-awareness use cases. It provides a working starting point for evaluation while clearly separating model-generated flags from verified findings.